

MATH 6610: Advanced Numerical Analysis**Homework 2**

Due: Friday, October 16, 2026

Instructions. Show all work. Numerical answers should be reported to the number of digits requested. You may use MATLAB or Python for the computational parts.

Problem 1. (Wallis's equation.) Apply Newton's method to $f(x) = x^3 - 2x - 5$ with starting value $x^0 = 2$.

- Compute x^1, x^2, x^3 , carrying at least ten digits.
- The root is $x^* = 2.0945514815 \dots$. Tabulate the errors $|x^k - x^*|$ and verify that the number of correct digits approximately doubles at each step.
- Using the error relation $|x^{k+1} - x^*| \approx C |x^k - x^*|^2$, estimate C from your table and compare with $|f''(x^*)/(2f'(x^*))|$.

Problem 2. Consider the nonlinear system

$$x_1^2 + x_2^2 = 1, \quad x_2 = x_1^3.$$

- Write the system as $f(x) = 0$ and compute the Jacobian $Df(x)$.
- Starting from $x^0 = (1, 1)^T$, carry out one full Newton step by hand: evaluate $f(x^0)$ and $Df(x^0)$, solve the 2×2 linear system for the correction Δx^0 exactly (fractions), and state x^1 .
- Without further computation, explain what linear system must be solved at the next step and which parts of the work from (b) can be reused.

Problem 3. (Simplified Newton method.) The simplified Newton iteration freezes the derivative at the starting point:

$$x^{k+1} = g(x^k), \quad g(x) = x - \frac{f(x)}{f'(x^0)} \quad (n = 1).$$

Let x^* be a zero of f with $f'(x^*) \neq 0$, and suppose f is continuously differentiable near x^* .

- Show that $g'(x^*) = 1 - f'(x^*)/f'(x^0)$.
- Using the mean value theorem, show that if $q := |1 - f'(x^*)/f'(x^0)| < 1$, then the iteration converges for all starting values sufficiently close to x^* , with $|x^{k+1} - x^*| \leq (q + \epsilon) |x^k - x^*|$ for any $\epsilon > 0$ eventually.
- Explain in one or two sentences why this method converges only linearly in general, and what one gains in exchange (think of the cost per step for systems).

Problem 4. Compute the condition number $\kappa(x)$ of the problem $F(x) = \tan x$ using the formula $\kappa(x) = |x| |F'(x)| / |F(x)|$.

- Show that $\kappa(x) = \left| \frac{2x}{\sin 2x} \right|$.
- Determine all points in $(0, \pi)$ near which the problem is ill-conditioned, and compute $\lim_{x \rightarrow 0^+} \kappa(x)$.

- (c) For $x = 1.5707$ (close to $\pi/2$), estimate $\kappa(x)$ and state how many digits of accuracy one can expect in $\tan x$ when x itself carries a relative error of 10^{-16} .

Problem 5. Consider evaluating

$$f(x) = \frac{1 - \cos x}{x^2}$$

for small x ; note $\lim_{x \rightarrow 0} f(x) = \frac{1}{2}$.

- (a) Evaluate the formula as written at $x = 10^{-8}$ in double precision (by hand reasoning or on a machine) and explain the result using the rounding of $\cos x$ and catastrophic cancellation.
- (b) Rewrite f using the identity $1 - \cos x = 2 \sin^2(x/2)$ and show that the new formula evaluates accurately for small x .
- (c) Which of the two computations is unstable, and is the underlying *problem* ill-conditioned at small x ? Justify with the condition number of f near 0.